Hw4

Wednesday, February 25, 2026 12:40 AM

4 Kinematic Equation

The kinematic equation relating the output position θ_2, the input position θ_1 and the misalignment angle α is as follows:

$$\theta_2 = \tan^{-1}\left(\frac{\tan(\theta_1)}{\cos(\alpha)}\right). \tag{1}$$

Assume that α is constant with respect to time.

1) $\frac{d\theta_1}{dt} = \omega_1 \qquad u = \frac{\tan(\theta_1)}{\cos\alpha}$

$$\omega_2 = \frac{d\theta_2}{dt} = \frac{1}{1+u^2} \cdot \frac{du}{dt}$$

$$\Rightarrow \omega_2 = \frac{1}{1+\left(\frac{\tan\theta_1}{\cos\alpha}\right)^2} \cdot \left(\frac{\sec^2(\theta_1)\ \omega_1}{\cos\alpha}\right) \cdot \frac{\cos^2\alpha\cos^2\theta_1}{\cos^2\alpha\cos\theta_1}$$

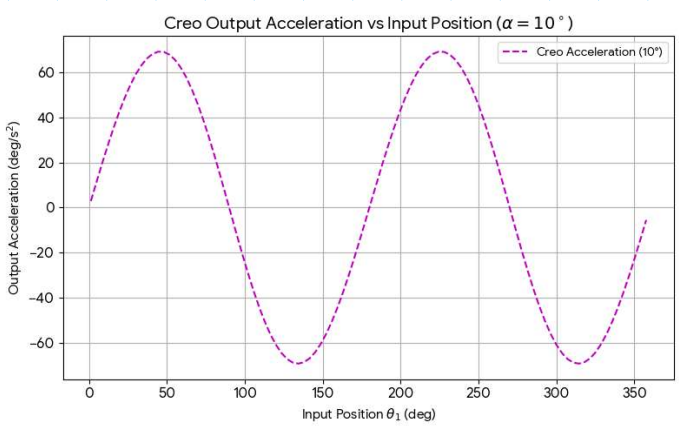
$$= \frac{\cos^2\alpha}{\cos^2\alpha + \tan^2\theta_1\cos^2\alpha} \cdot \frac{\omega_1}{\cos\alpha\cos^2\theta_1}$$

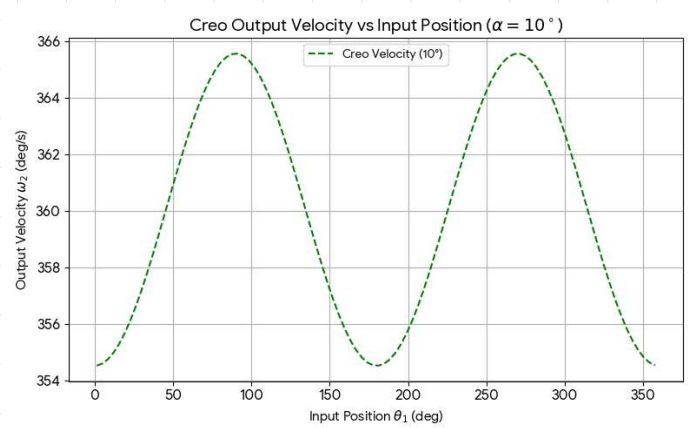
$$= \frac{\omega_1\cos\alpha}{\cos^2\alpha\cos^2\theta_1 + \sin^2\theta_1} \qquad \text{recall } \cos^2\alpha = 1-\sin^2\alpha$$

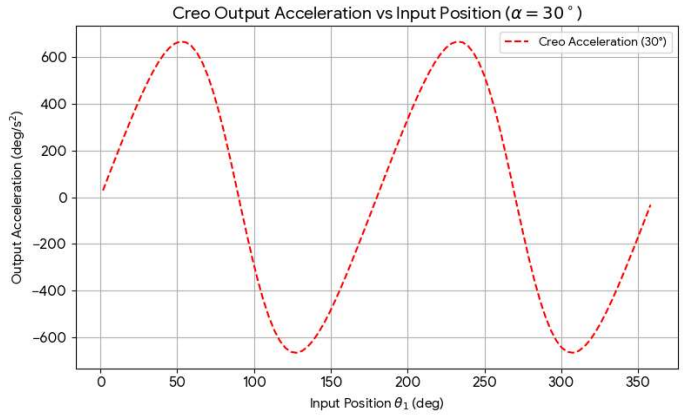
$$= \frac{\omega_1\cos\alpha}{(1-\sin^2\alpha)\cos^2\theta_1 + \sin^2\theta_1}$$

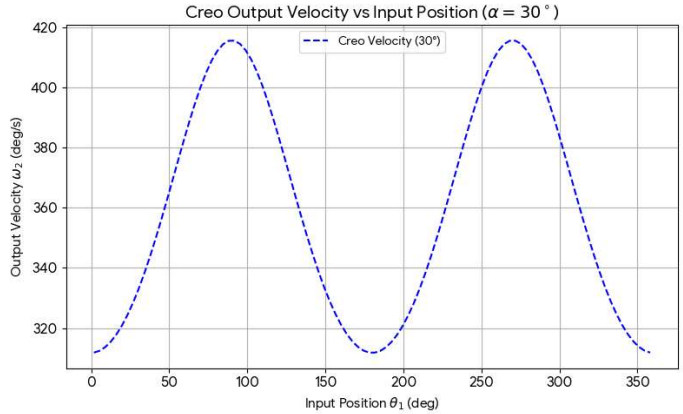
$$= \frac{\omega_1\cos\alpha}{\cos^2\theta_1 - \sin^2\alpha\cos\theta_1 + \sin^2\theta_1} \Rightarrow \frac{\omega_1\cos\alpha}{1-\sin^2\alpha\cos^2\theta_1}$$

2)









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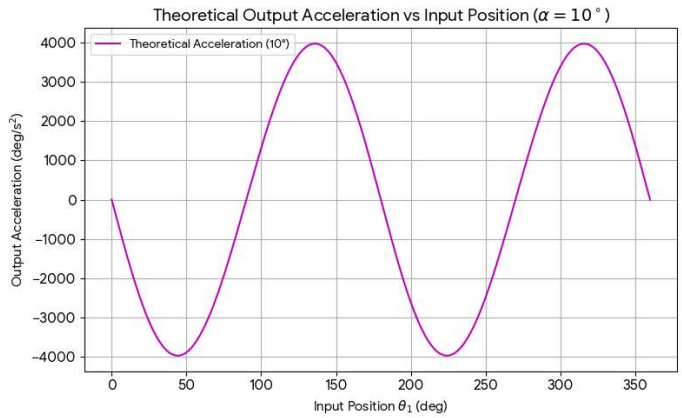

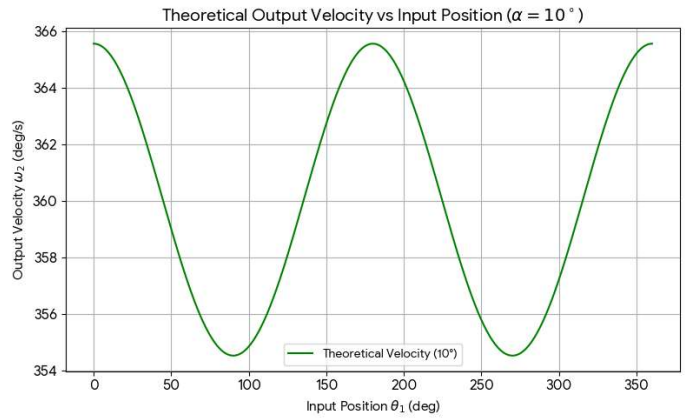

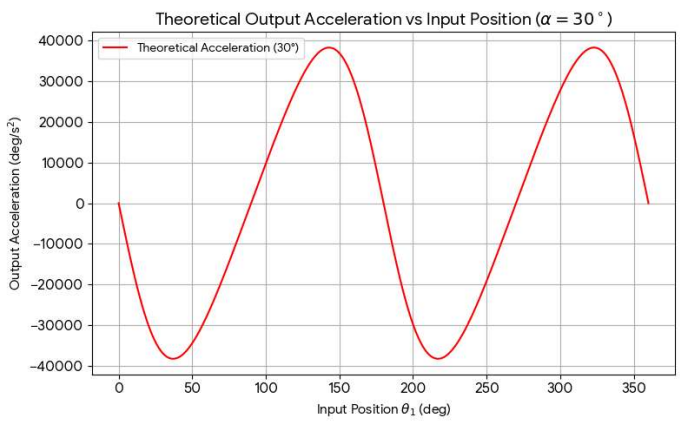

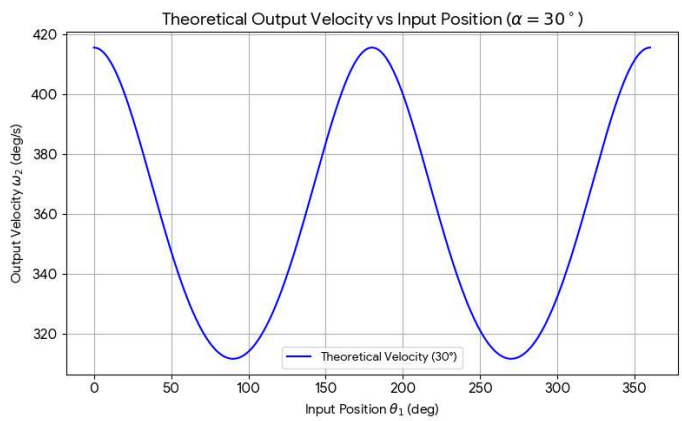

4) Brief explanation of results

The curves are identical as motion transfer through the assembly reflect the plot when they are overlayed. The misalignment can cause the shaft to fluctuate, speeding up and slowing down twice per revolution. So as misalignment increases, amplitude increases as well:

- At 10 degrees, velocity begins to fluctuate while acceleration remains relatively low around ± 35 deg/sec^2
- At 30 degrees, velocity fluctuations increase. Because speed changes violently, acceleration increases nearing ± 800 deg/sec^2